COMPUTER NETWORKS - LECTURE 1

Topic: OSI Reference Model Architecture

The Open Systems Interconnection model defines seven abstraction layers:

1. Physical Layer: Transmission of raw bitstream over physical media
2. Data Link Layer: Node-to-node data transfer and error detection (Ethernet, MAC)
3. Network Layer: Packet forwarding and routing through intermediate routers (IP, ICMP)
4. Transport Layer: End-to-end communication and reliability (TCP, UDP)
5. Session Layer: Dialogue control and token management
6. Presentation Layer: Data formatting, encryption, and compression
7. Application Layer: End-user protocols (HTTP, DNS, FTP, SMTP)

COMPUTER NETWORKS - LECTURE 2

Topic: Transport Layer Protocols

Transmission Control Protocol (TCP) provides connection-oriented, reliable transmission.

User Datagram Protocol (UDP) provides connectionless, low-latency transmission.

TCP three-way handshake: SYN -> SYN-ACK -> ACK.